Yinghong(Iris) Wang

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Education

University of Illinois at Urbana-Champaign **September 2021-May 2025**

- Bachelor of Science; Major in Computer Science and Economics; GPA:3.87/4.0
- Relevant Coursework: *Applied Machine Learning(Python), Database Systems(SQL), Data Structures(C++), Intro to Computer System(C/Python), Programming Languages and Compilers(Haskell), Intro to Algorithms & Models of Computer*

Northwestern University **Expected December 2026**

- Master of Science in Machine Learning and Data Science; GPA: 3.79/4.0

Work Experience

IDX Exchange, Agentic AI Engineer Intern (https://github.com/yinghongwang02/idx-agentic-ai-nlp-project)

Real Estate Agentic Copilot | Python, LangGraph, Streamlit, MySQL, OpenAI API, FAISS, Docker **June 2026 – Sep. 2026**

- Built a **LangGraph multi-agent real estate copilot** orchestrating agents for **property search, market analysis, recommendation, compliance, and document-grounded RAG**, with intent-based routing, shared workflow state, and mixed-intent execution.
- Developed and evaluated **structured, semantic, and hybrid retrieval** across **53K+ active listings, 87K+ sold records, and 52K+ embeddings**, combining **parameterized MySQL filtering, FAISS vector retrieval, and hybrid similarity ranking**.
- Built an **explainable recommendation and market-analysis pipeline** using sold comparables, preference matching, comparable-value analysis, and negotiation signals; optimized bounded parallel execution to achieve **2.2× speedup and 54.7% lower latency**.
- Integrated real-world agent actions through **OpenClaw, WhatsApp, and Gmail**, implementing **human-in-the-loop approval and outbound safety checks** to prevent unauthorized external communication.
- Built **retrieval and LLM-as-a-Judge evaluation** for semantic search and RAG quality; containerized and deployed the application with **Docker and Streamlit Cloud** with a 268-test regression suite.

HeliosIQ Technologies, Inc., Software Engineering Intern

Healthcare Rating Prediction System | XGBoost, Feature Engineering, PostgreSQL **June 2026 – Sep. 2026**

- Built an **end-to-end ML pipeline** for healthcare performance prediction, integrating CMS data with temporal feature engineering.
- Developed reusable **PostgreSQL and Python ETL workflows** for healthcare data preparation, feature engineering, and chronological train/validation/test splits.
- Integrated **a multi-stage predictive modeling workflow** using Ridge Regression and XGBoost to analyze changes in healthcare performance indicators.
- Built reproducible model training and evaluation workflows with temporal validation, feature ablation, and metrics including MAE, RMSE, and R^2.

Projects & Research

Sabre Labs, Capstone: Agentic AI Hotel Recommendation System | Multi-Agent Systems **Sep. 2026 – Present**

- Collaborated on an **agentic hotel recommendation system** integrating synthetic travelers, booking history, hotel search, and choice modeling.
- Built a **persistent synthetic traveler framework** with 200 configurable profiles, separating long-term preferences from booking history.
- Implemented **anonymous/returning user identity controls** and reproducible run-level sampling with configurable user ratios and deterministic seeds.
- Developed **cross-experiment history persistence** with profile reloading, incremental updates, and duplicate-request prevention, validated through automated tests.

Northwestern University, Advanced RAG System for Environment News Analysis | LangChain, LangGraph, ChromaDB, LoRA
(https://github.com/yinghongwang02/Advanced-RAG-System-for-Environment-News-Analysis) **Jan. 2026-Mar. 2026**

- Designed a **LangGraph-based multi-agent RAG pipeline** for environmental news question answering, integrating query rewriting, vector retrieval, document relevance grading, and grounded answer generation.
- Built a persistent **ChromaDB vector store** using OpenAI embeddings, processing up to **300 Guardian environmental news articles**.
- Implemented **LLM-based document relevance grading**, and fine-tuned **SmolLM2-360M with LoRA** for relevance classification.
- Developed an **evaluation framework** comparing **Base LLM, Basic RAG, Advanced RAG with base LLM grading, and LoRA-enhanced RAG**, analyzing retrieval quality, answer grounding, and the precision–coverage trade-off.

Global IQ, Practicum: Machine Learning Consultant, Mobility Cost Prediction System |AWS, LightGBM **Sep. 2025-May 2026**

- Collaborated on developing scalable **ML pipelines** on **AWS** to predict global mobility costs from enterprise relocation datasets.
- Performed large-scale data cleaning and feature engineering on enterprise mobility datasets spanning relocation, taxation, and compensation domains.
- Benchmarked **Linear Regression, SVM, Neural Networks, Random Forest, and LightGBM models**, evaluating predictive performance across salary, taxation, and relocation cost targets.
- Contributed to an **LLM-assisted feature generation workflow** to transform unstructured domain-specific information into structured features for downstream ML models.

Northwestern University, CTA Real-Time Train Tracking System | Python, FastAPI **Nov. 2025-Dec. 2025**

- Collaborated on a **full-stack** system for real-time train tracking in Chicago using the CTA API, updating positions every 30 seconds.
- Extended **FastAPI backend** and Python data pipeline for real-time data ingestion and **SQLite** storage.
- Worked with an interactive map (**Leaflet.js**) with route filtering and snapping logic (Turf.js) to align train positions with tracks.
- Conducted EDA across all Chicago CTA train lines, examining route-level service patterns to support real-time train tracking.

Technical Skills

Languages: Python, SQL, C++, Java, C, R, Haskell, Julia

AI / Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, OpenAI API, LangChain, LangGraph, ChromaDB, Agentic AI, Multi-Agent Systems, RAG, LoRA, LLM Evaluation, Semantic Search, FAISS, Feature Engineering, Temporal Validation

Backend / Data Engineering: FastAPI, Flask, Streamlit, REST APIs, MySQL, PostgreSQL, SQLite, ETL, Data Pipelines

Cloud / Systems: AWS , Docker, Linux, Git